KRISHNENDU GHOSH

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Bachelor of Technology - Computer Science (Artificial Intelligence and Machine Learning), 2027
Supreme Knowledge Foundation Group of Institutions (Affiliated to MAKAUT ,WB) – Hooghly, WB
• Current GPA 7.22(Upto 4th Semester)Average of all Sem

Higher Secondary Education: PCM & CS, 2023
Uttarpara Amarendra Vidyapith - Hooghly, WB
• WBCHSE Board • Percentage: 61.6%

Secondary Education, 2021
Uttarpara Amarendra Vidyapith - Hooghly, WB
• WBBSE Board • Percentage: 79.71%

Technology & Skills

Programming Languages: Python, Java, C/C++, JavaScript, Dart, SQL
AI/ML & Data Science: Machine Learning, Artificial Intelligence, Computer Vision, Transfer Learning, CNN, LSTM, Data Analysis, Pattern Recognition, Explainable AI (Grad-CAM, SHAP), LangChain (GPT-4, GPT-3.5)
Web & Mobile Development: React.js, Flutter, HTML, CSS, REST API Development, FastAPI, Streamlit, Android Studio
Databases & Cloud: Firebase Firestore, Firebase Authentication, MySQL, PostgreSQL, SQLite
Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, MATLAB
Tools & Platforms: Git, Google Colab, Google Earth Engine, Linux, Render (deployment), Vercel ,Nectlify
Core Computer Science: Data Structures & Algorithms, Operating Systems, Database Management Systems

Projects

AgriVision – AI-Driven Precision Agriculture Monitoring Platform-Android App **Sept 2025**
Tech: Python, FastAPI, Flutter, Firebase, MATLAB, CNN, LSTM, Google Earth Engine, OpenWeather API

- Built a precision agriculture system leveraging hyperspectral imaging, environmental sensors, and weather APIs, with CNN/LSTM models for crop health, vegetation stress, and pest risk prediction.
- Developed a FastAPI backend and Flutter mobile dashboard integrated with Firebase to deliver real-time farm insights using spectral data and soil/weather inputs.

Clean Route – Al-Enabled Smart Waste Management and Route Optimization System – Android APP
Tech: Flutter, Firebase, Python, REST API **Nov,2025**

- Developed a cross-platform Flutter application with Firebase integration for real-time waste collection monitoring, including authentication and database management.
- Implemented Python-based backend APIs for route optimization and efficient waste collection planning.

TextSage AI – Intelligent Text Summarization Tool*(Web App)* Live- *https://textsage.krishnendughosh.in/*
Tech: React.js, Tailwind CSS, Python,Node.js, FastAPI, REST API **Feb, 2026**

- Developed an AI-powered NLP platform for summarization, sentiment analysis, and classification by implementing pipelines for keyword extraction, entity recognition, and semantic analysis.
- Built scalable backend APIs for real-time processing and a responsive React frontend, optimizing preprocessing and inference to support applications like document summarization, review analysis, and research assistance.

OncoScan XAI – Explainable Multi-Modal Transfer Learning Framework for AI-Assisted Breast Cancer Detection **[Currently Working]**
Tech: Python, PyTorch, Radiomics, ResNet50, EfficientNet-B0,Scikit-learn, FastAPI, React.js, Firebase , Grad-CAM, SHAP, REST API

- Developed an AI-assisted breast cancer detection system using multi-modal MRI and mammography data, implementing transfer learning (ResNet50, EfficientNet) with feature fusion to improve classification performance.
- Integrated Explainable AI techniques (Grad-CAM, SHAP) for interpretability, and built an end-to-end pipeline including preprocessing, FastAPI-based inference APIs, React frontend for visualization/reporting, and Firebase for secure data management.

Disclaimer

I Solemnly declare that all the above information is correct to the best of my knowledge & belief.

Date 15/04/2026